

One Electron at a Time

KKKC6353 Nanoelectronic Devices · Lecture 3, Part 3 of 3 · Coulomb blockade, single-electron transistors and Josephson junctions

Dr Muhammad Faiz Bukhori

Universiti Kebangsaan Malaysia · 2026

1

Charge comes in lumps

One more electron on a tiny island costs real energy.

A crowded lift

Whole electrons

Charge always comes in whole electrons. You cannot add half an electron to anything.

Big object, no cost

One more person in a goods lift makes no difference; one more electron on a large piece of metal costs almost nothing.

Tiny island, real cost

In a full tiny lift, one more person is a squeeze. On a nanometre island, one more electron costs tens to hundreds of meV.

The charging energy against island size

- Charging energy **$e^2/2C$**: smaller island, smaller C, **larger cost**
- SiO_2-embedded sphere: **100 nm → 3.7 meV**, **10 nm → 37 meV**, **2 nm → 185 meV**
- Thermal energy: **26 meV** at 300 K, 0.36 meV at 4.2 K
- Clean blockade needs about **10 × k_BT**: islands of **1–2 nm** at room temperature

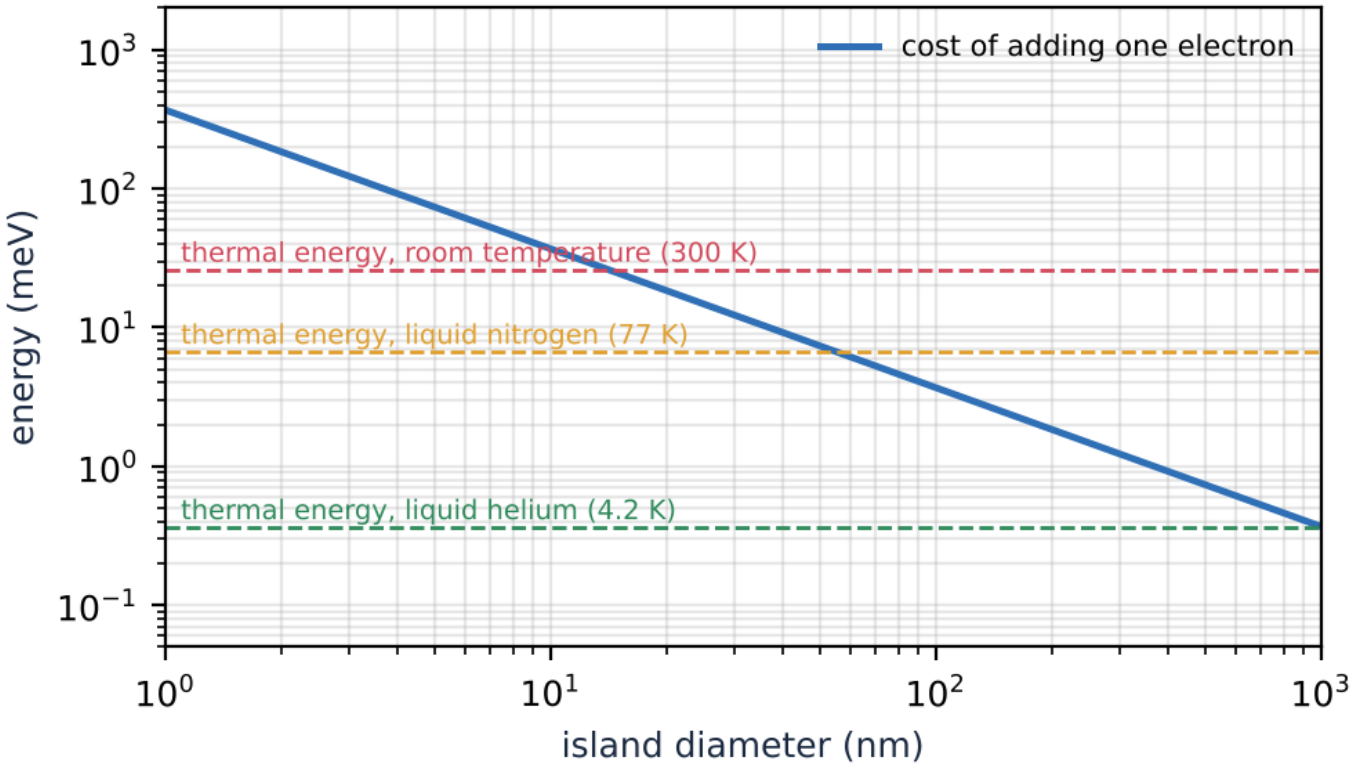


Figure made for this course.

2

Coulomb blockade

Blocked by the repulsion of a single extra charge.

Blocked, and the two conditions

Blocked

An island between source and drain, joined by two tunnel junctions. If the voltage cannot pay the charging energy, no electron can tunnel on, so none can tunnel off.

1 · Beat the heat

The charging energy must be well above $k_B T$, or thermal kicks let electrons through anyway: a few nm at 300 K, or below ~1 K for a micrometre.

2 · Opaque junctions

Each junction needs more than about 26 kΩ (h/e^2), so each electron really sits on the island before it leaves.

3

The single-electron transistor

A gate that adds one electron at a time.

The structure

- A MOSFET whose channel is replaced by a **tiny island**
- **Two tunnel junctions**: source → island → drain
- A **gate** close to the island; no electrons cross to it
- The gate only changes the island's **energy**

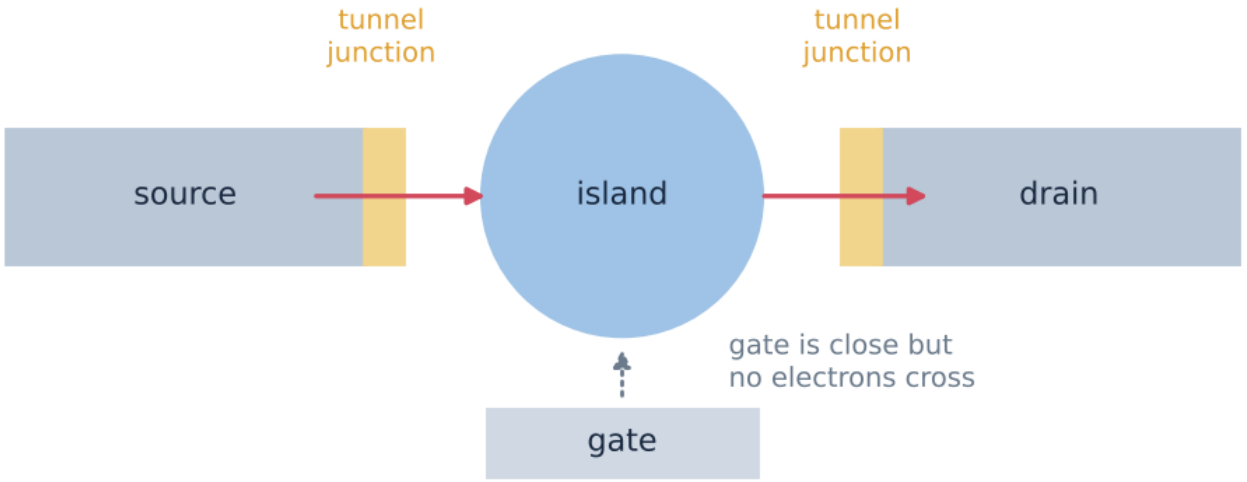
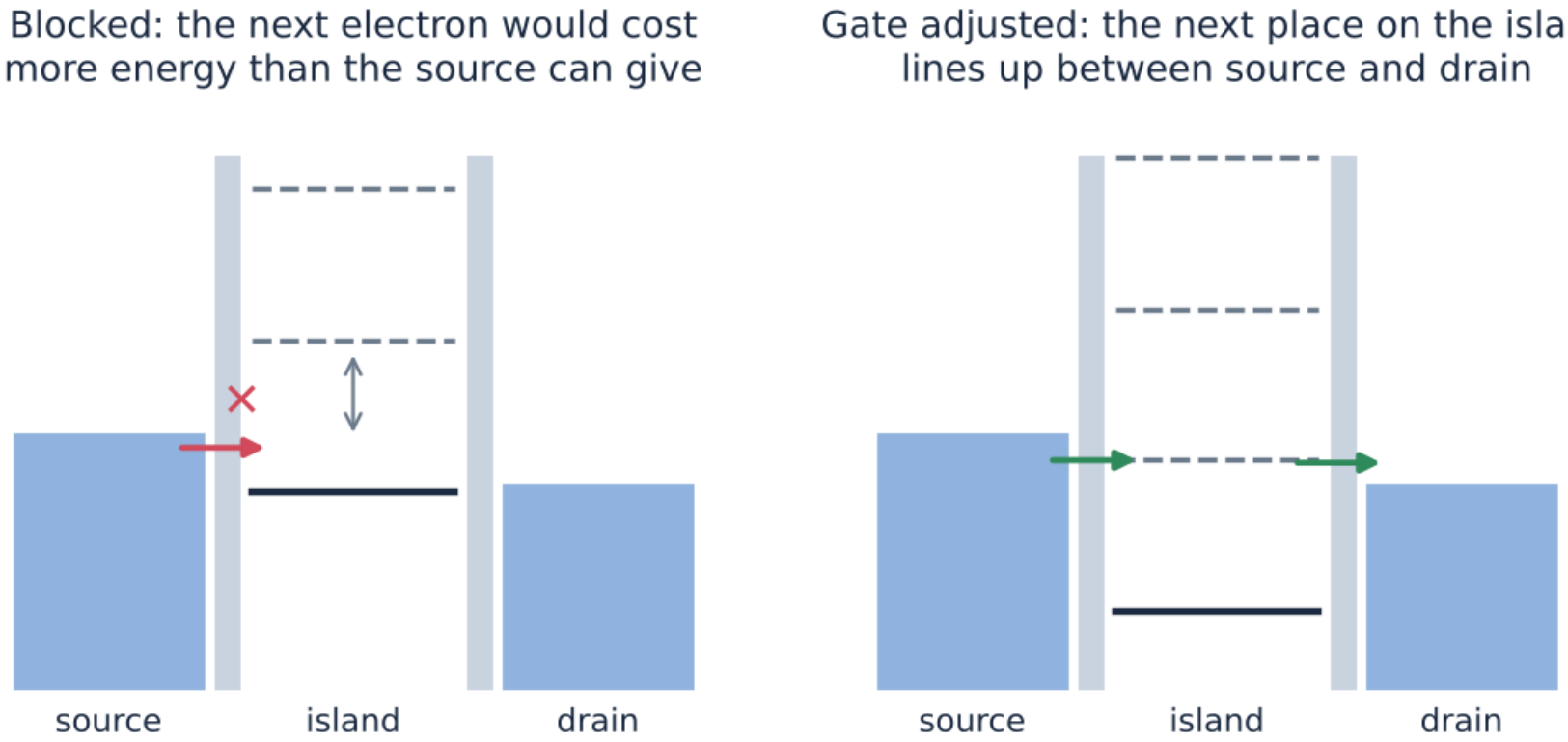


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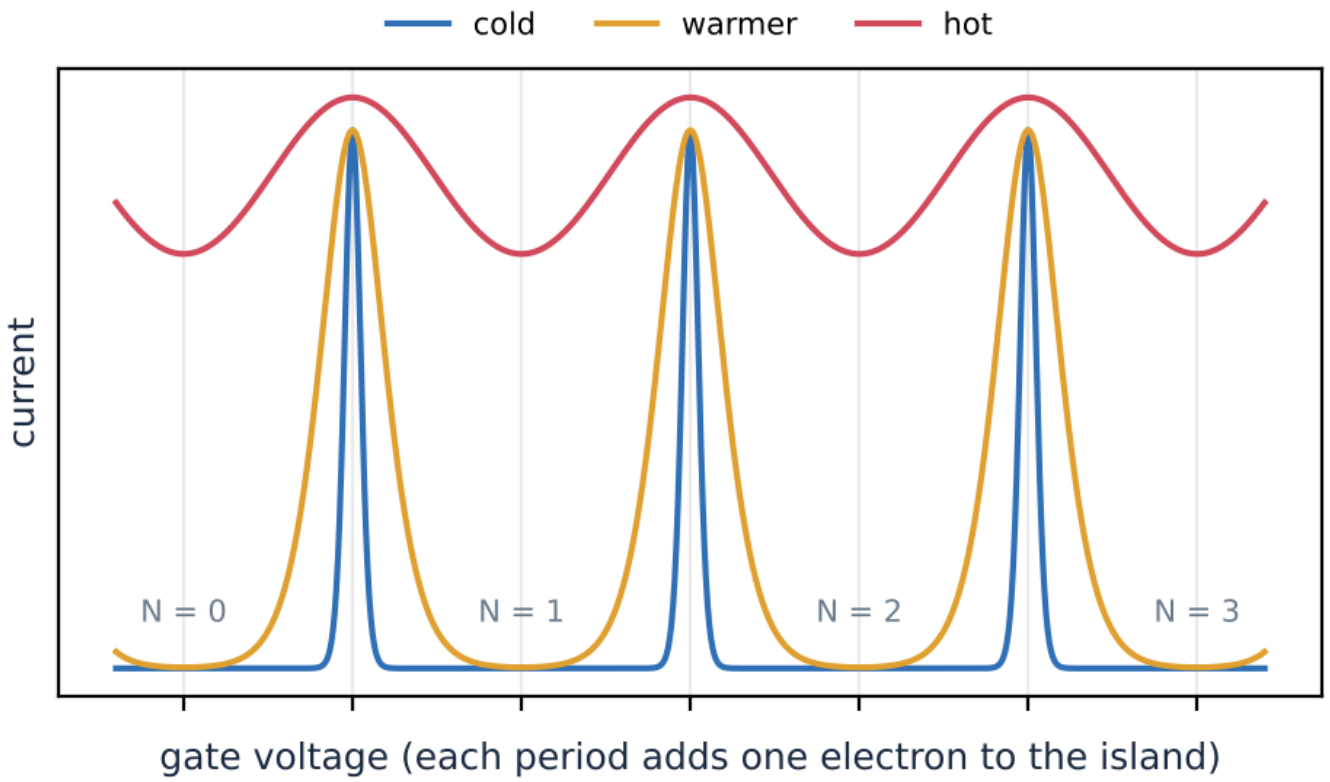
How the gate controls it



Schematic. Figure made for this course.

The island's places form a ladder, one charging energy apart. **Blocked** when the next empty place is above the source. The gate lowers the ladder until a place sits **between source and drain**: electrons pass **one at a time**.

Coulomb oscillations

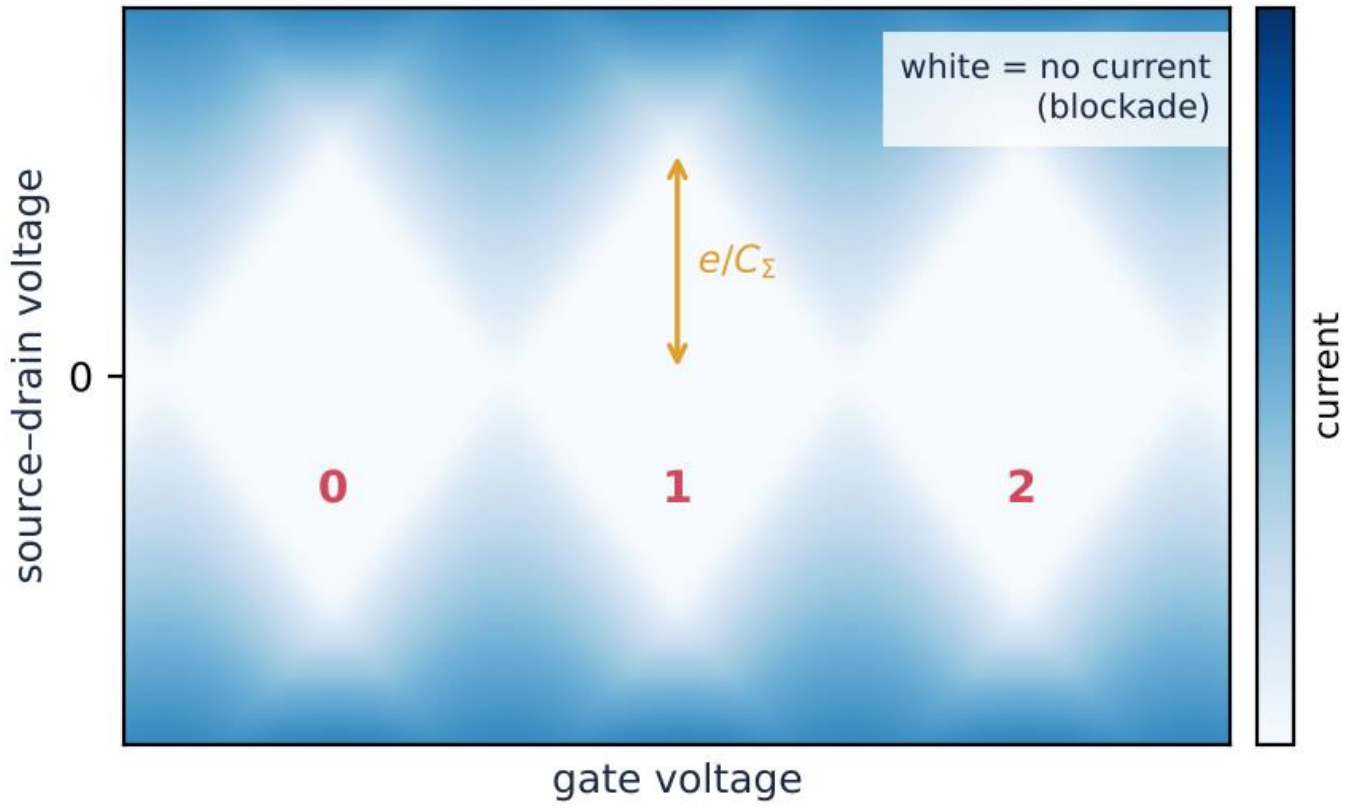


Calculated with the orthodox theory. Figure made for this course.

The current switches on and off **again and again**: one peak per extra electron (period e/C_g). Heat blurs the peaks: at 40% of the charging energy they have almost gone.

Coulomb diamonds

- Two knobs: **gate** and **source–drain** voltage
- Inside each white **diamond**: a fixed number of electrons, **no current**
- Diamond height $\mathbf{e/C_\Sigma}$ measures the island's capacitance
- Outside: current flows



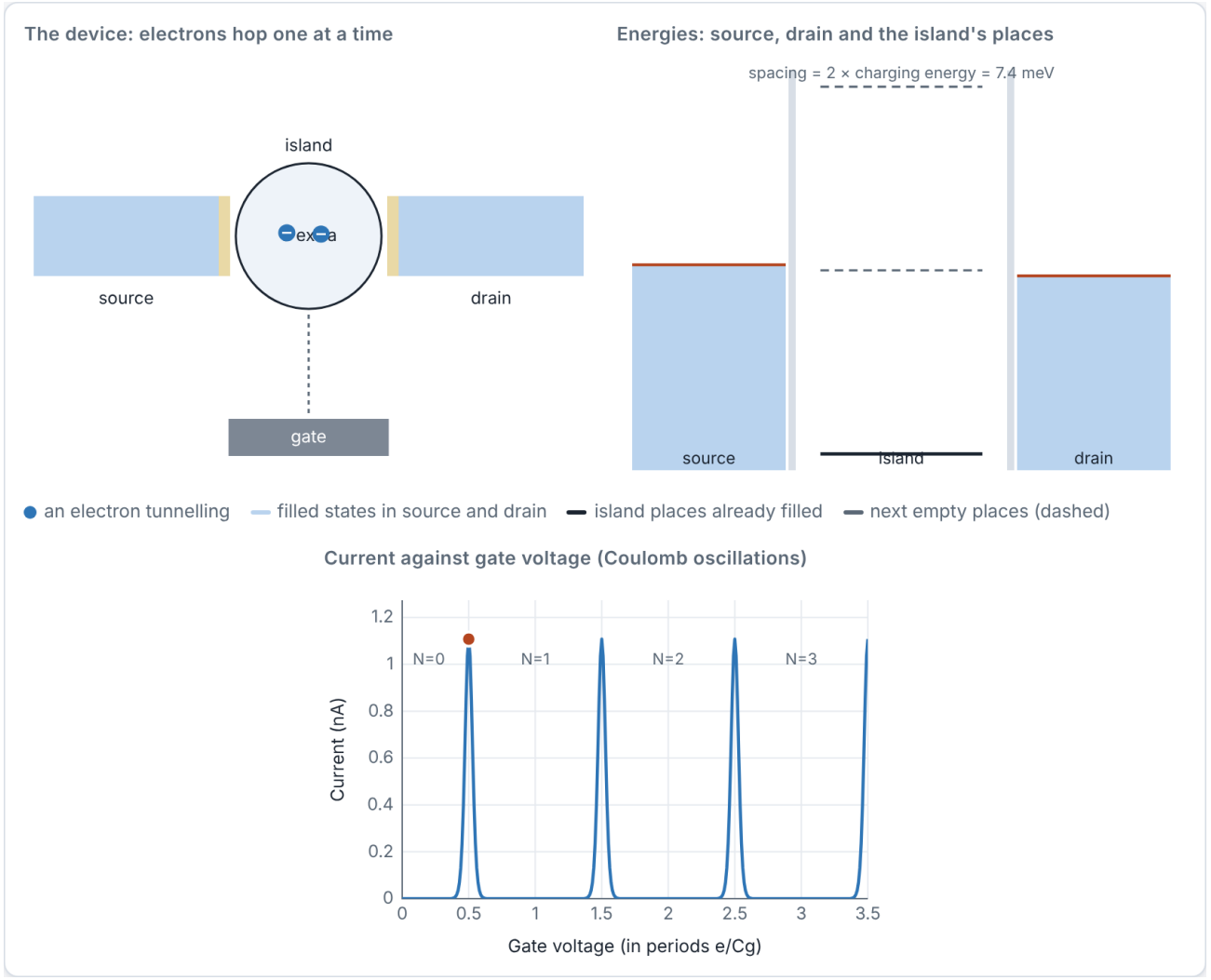
Calculated with the orthodox theory. Figure made for this course.

Interactive animation 6

Single-electron transistor

faizverse.github.io/nanoelectronic-devices/L3/single-electron-transistor.html

- From a blocked valley, raise the gate until current flows
- Count the extra electrons after each peak
- Warm the 100 nm island from 1 K to 30 K
- Shrink it to 2 nm at 300 K: are the peaks still there?



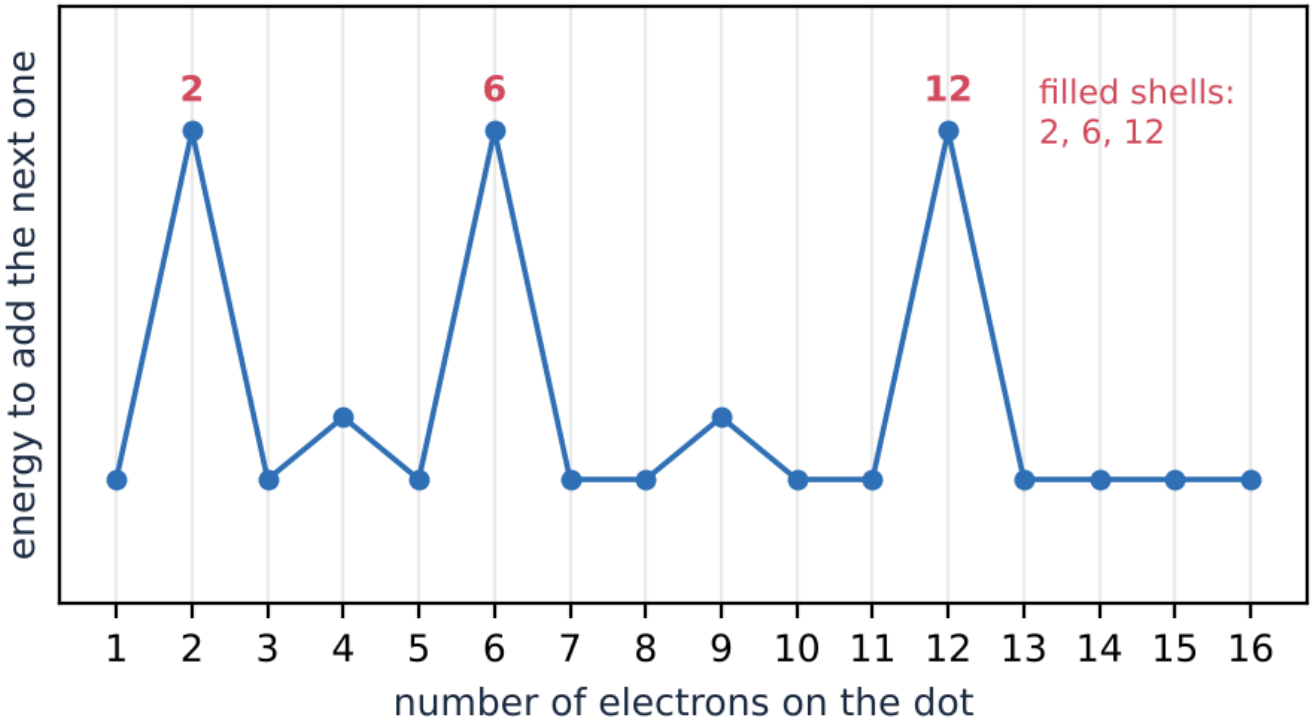
4

Artificial atoms

A small quantum dot fills shells like an atom.

Shells in a quantum dot

- A semiconductor dot has **quantised levels** (Lecture 2)
- Adding an electron costs the charging energy, plus a jump when a **level is full**
- Tarucha et al., 1996: extra cost after **2, 6 and 12** electrons
- Smaller bumps at 4 and 9: **Hund's rule**, as in real atoms
- Size, shape and 'chemistry' **set by design**



Schematic, after Tarucha et al. (1996). Figure made for this course.

5

What single-electron devices are good for

Not computer chips, but the most sensitive charge detectors ever made.

Not a replacement for the MOSFET

1–2 nm islands

Room temperature needs atomically precise islands, by the billion. Shown in the lab, not in production.

Tiny current

A few nanoamperes through two high-resistance junctions: slow circuits that cannot drive wires.

Too sensitive

One stray charge nearby shifts the oscillations, and each SET on a chip would shift differently.

Sensing and counting

The most sensitive electrometer

Biased halfway up a Coulomb peak, a fraction of one electron's charge changes the current a lot. Silicon spin-qubit processors read each qubit's spin, converted to charge, with a single-electron sensor.

A new ampere

A single-electron pump moves exactly one electron per cycle: $I = ef$, about 0.16 nA at 1 GHz, accurate to about one part in a million. Since 2019 the electron charge is exact: $1.602\,176\,634 \times 10^{-19}$ C.

The quantum metrology triangle

- **Current**: electron pump, I = ef
- **Voltage**: Josephson effect, V = (h/2e) f
- **Resistance**: quantum Hall effect, R = h/e²
- Ohm's law links all three, so the triangle **tests** our understanding of each

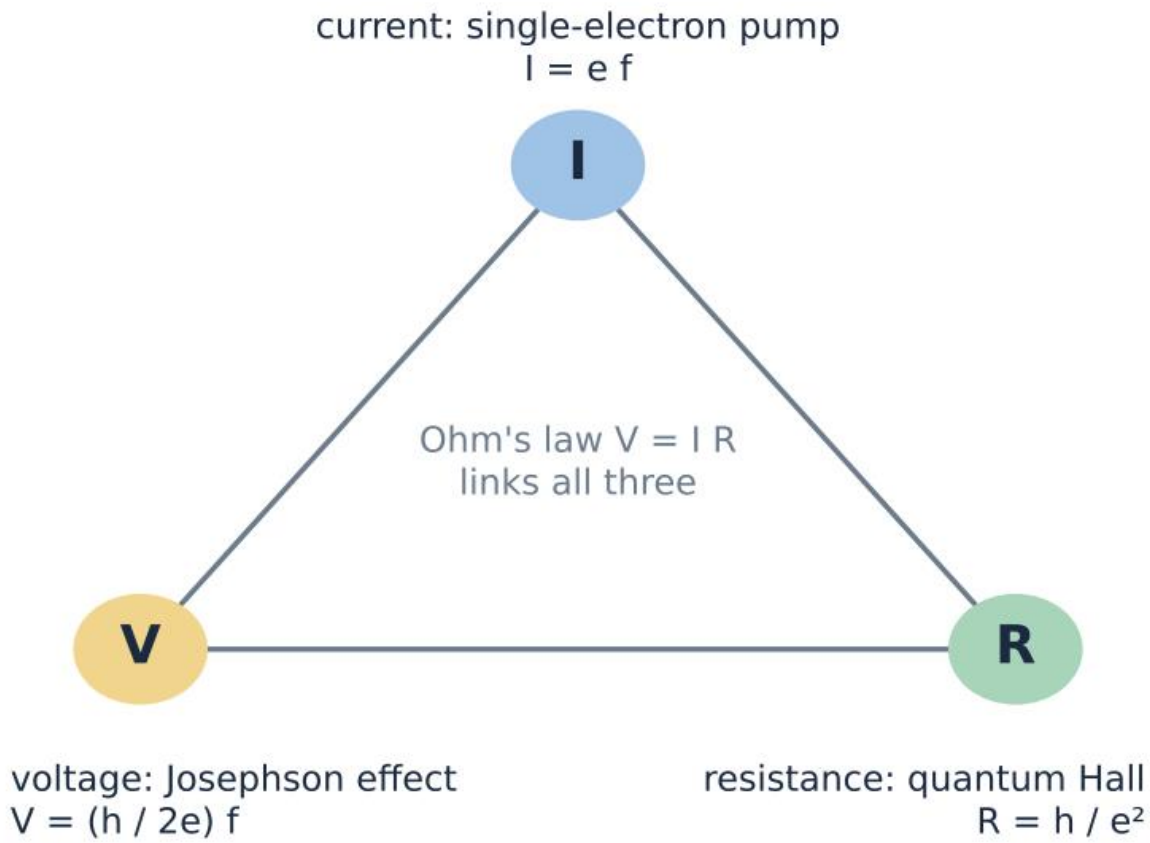


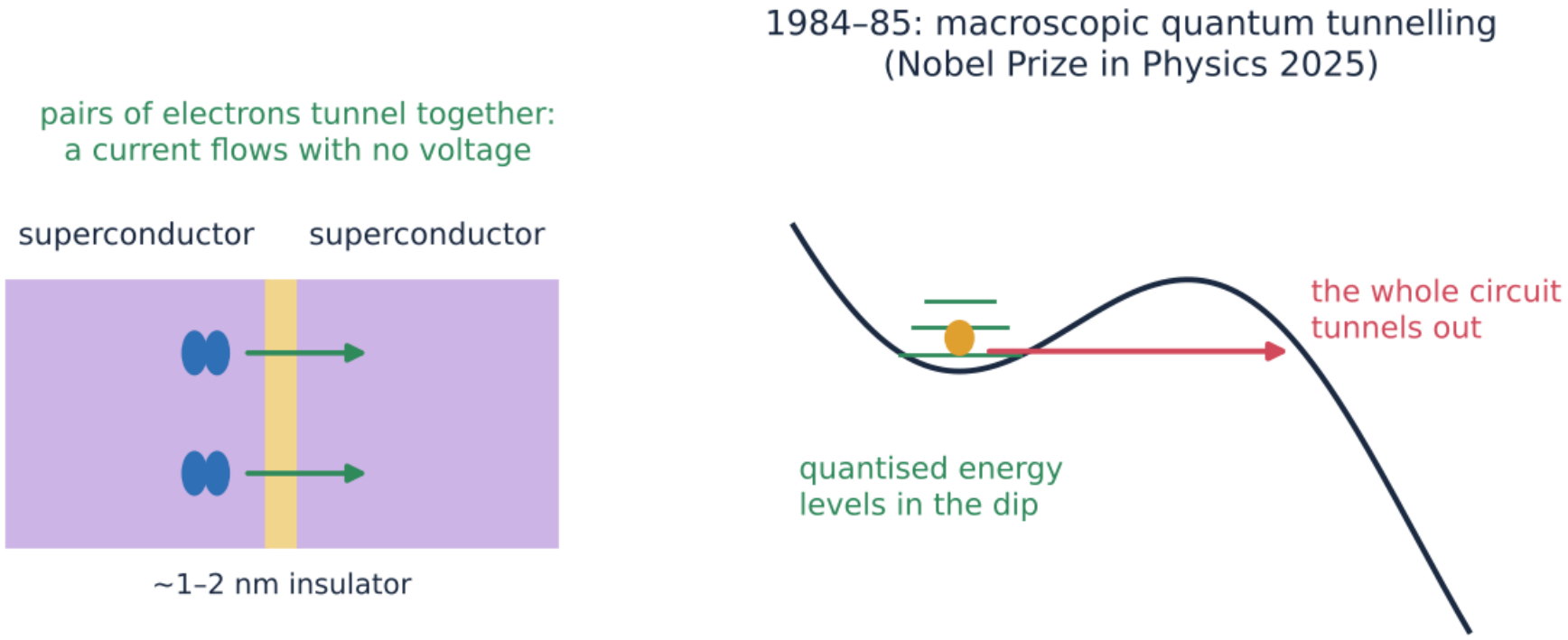
Figure made for this course.

6

Tunnelling pairs: the Josephson junction

A whole circuit tunnelling as one quantum object.

The Josephson effect



Schematic. Figure made for this course.

In a superconductor electrons form **Cooper pairs**. Pairs tunnel through a 1–2 nm insulator and carry current with **no voltage** (Josephson, 1962; Nobel 1973). At voltage V the current oscillates at 2eV/h, **484 GHz per mV**.

The 2025 Nobel Prize in Physics

Berkeley, 1984–85

John Clarke, Michel Devoret and John Martinis biased a Josephson junction just below the current at which a voltage appears.

Macroscopic tunnelling

At the lowest temperatures the whole circuit still escaped its energy dip, at a rate independent of temperature: it tunnelled. Microwaves revealed quantised levels.

Qubits

A circuit with quantised levels can be a qubit. Google's and IBM's superconducting processors are built from such circuits.

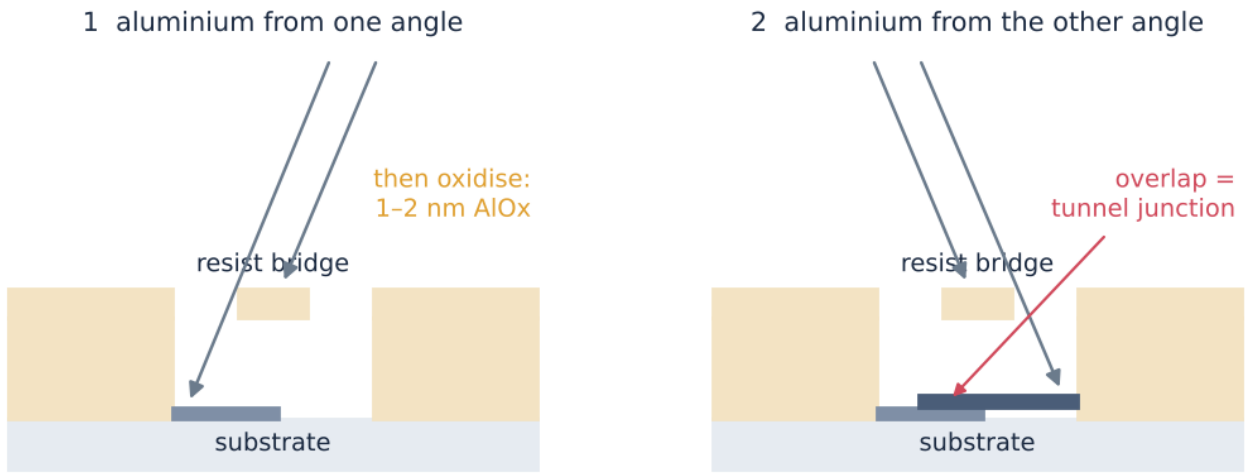
7

How single-electron devices are made

Two angles, one tiny overlap.

Shadow evaporation and other routes

- **Dolan bridge**: aluminium evaporated from one angle, oxidised, then from another
- The two layers **overlap** under the bridge, with 1–2 nm of AlO_x between: a tunnel junction
- Still used for most **superconducting qubits**
- **Silicon**: islands and barriers defined by gate voltages, on 300 mm lines
- **Nanoparticles** a few nm across for room temperature



Schematic. Figure made for this course.

Preview: three transistors compared

	MOSFET	Single-electron transistor	Carbon nanotube FET
Channel	Thin layer of silicon	Tiny island, two tunnel junctions	Semiconducting nanotubes
How electrons travel	Over a gate-set barrier	On and off the island, one at a time	Along the tube, often without scattering
Current vs gate	Turns on once	Peaks repeatedly	Turns on once
Room temperature?	Yes	Only with 1–2 nm islands	Yes
Status in 2026	Every computer chip	Sensors, qubit readout, metrology	Research prototypes (16-bit processor, 2019)

Part 3 in seven points

1 Lumps of charge

One electron on a tiny island costs $e^2/2C$.

2 Blockade

Needs small or cold islands and junctions above 26 kΩ.

3 The SET

A gate shifts the island: one current peak per electron.

4 Artificial atoms

Small dots fill shells of 2, 6 and 12.

5 Uses

Charge sensing, qubit readout, defining the ampere.

6 Josephson

Cooper pairs tunnel: voltage standards and SQUIDs.

7 2025 Nobel

A whole circuit tunnels: the basis of superconducting qubits.

Credits

Made for this course

All figures were made for KKKC6353 (scripts in figures-source/); Coulomb oscillations and diamonds are orthodox-theory calculations. Animation screenshot from faizverse.github.io/nanoelectronic-devices. Sections 1–3 draw on F. J. Himpsel's SET slides (UW–Madison), used in earlier years and now archived.

Sources

Fulton and Dolan, PRL 59, 109 (1987); Tarucha et al., PRL 77, 3613 (1996); Josephson, Phys. Lett. 1, 251 (1962); Devoret, Martinis and Clarke, PRL 55, 1908 (1985); Martinis, Devoret and Clarke, PRL 55, 1543 (1985); Likharev, Proc. IEEE 87, 606 (1999); Pekola et al., RMP 85, 1421 (2013). Video: NTT, 'Silicon single-electron devices'.